

MECHANICAL ENGINEERING

2026 & 2027 OFFICIAL RATING FORM

MIDDLE SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

☐ Entry passes compliance checklist

☐ ENTRY NOT EVALUATED

PRELIMINARY ROUND – SCORING (40 points)

1st: 40 Points	2nd: 39 Points	3rd: 38 Points	4th: 37 Points	5th: 36 Points	6th: 35 Points	7th: 34 Points
8th: 33 Points	9th: 32 Points	10th: 31 Points	11th: 30 Points	12th: 29 Points	13th or greater: 28 Points	
PRELIMINARY ROUND – SCORING SUBTOTAL (40 points)						

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right.

Indicate the rule violated: _____

PRELIMINARY SUBTOTAL (40 points)

SEMIFINAL – DOCUMENTATION PORTFOLIO (60 points)

CRITERIA	Minimal performance	Adequate performance	Exemplary performance
	1-4 points	5-8 points	9-10 points
Portfolio (X1)	Portfolio is unorganized and/or is missing three (3) or more components.	Portfolio is missing one (1) or two (2) components and/or is loosely organized.	Portfolio has all required components in order and is well organized.
Engineering Design Process (X1)	The description of the engineering design process is unclear, unconvincing, and lacks project details.	The description of the engineering design process includes some project details and explanation of steps in the design process.	The description of the engineering design process includes a full explanation of each step in the process; the description is defined, and detailed
Bill of Materials (X1)	Bill of materials is included, but more than one (1) material is missing.	Bill of materials is included with one (1) material missing; List is generally organized.	Bill of materials is included; it includes all materials, components, quantities, and is well organized.

Record scores
in the column
spaces below.

SEMIFINAL – DOCUMENTATION PORTFOLIO (60 points) – continued					
Design Sketches and Drawings (X1)	Only one (1) drawing/sketch of the model is included with little to no labeling.	Two (2) or more drawings/sketches of the model are included and are labeled.	Detailed drawings/sketches of the entire device/model are provided and appropriately labeled		
Work Log (X1)	The Work Log lacks major elements of the project.	The Work Log is somewhat complete, and generally reflects the time and work necessary for the project	The Work Log completely and accurately reflects the time and work necessary for the projects.		
References and Resources (X1)	There is little to no effort to provide references and resources.	References and resources are included and are somewhat credible.	There is clear evidence of the appropriate use of applicable references and resources.		
SEMIFINAL – DOCUMENTATION PORTFOLIO SUBTOTAL (60 points)					

SEMIFINAL ROUND – HEAD-TO-HEAD SCORE (20 points)						
1st: 20 Points	2nd: 18 Points	3rd: 16 Points	4th: 14 Points	5th-8th: 12 Points	9th-12th: 10 Points	
SEMIFINAL ROUND – HEAD-TO-HEAD SCORE SUBTOTAL (20 points)						

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____	
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SEMIFINAL SUBTOTAL (80 points)	
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To arrive at the TOTAL score, add any subtotals and subtract rules violation points, as necessary.	TOTAL (120 points)	
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Comments: I certify these results to be true and accurate to the best of my knowledge. JUDGE Printed name: _____ Signature: _____

CATAPULT SCORE SHEET

A shot attempt points may be modified if it is affected by a later shot

Participant/Team ID# _____

CATAPULT COMPLIANCE CHECKLIST		YES	NO
Does the team have safety goggles?			
Can the catapult be weighed down with two (2) Olympic weight plates (two [2] 25lbs)?			
Is the catapult within the size specifications?			
Is the catapult built with the correct materials?			
Does the catapult launch with a pull cord with at least five (5) feet of safety distance?			
Does the catapult have a safe launching mechanism?			
Is the catapult safe to operate?			
Please mark an "X" for each criteria. A "No" to any of the requirements will not advance.			

Shot Attempt #	Missed Shot 0 Points	Floor Zone 1 Point	On the Board 3 Points	In the Hole 5 Points	Modified Points
#1					
#2					
#3					
#4					
#5					
#6					
#7					
#8					

Total Points Scored:	
Time Remaining: (MM:SS)	